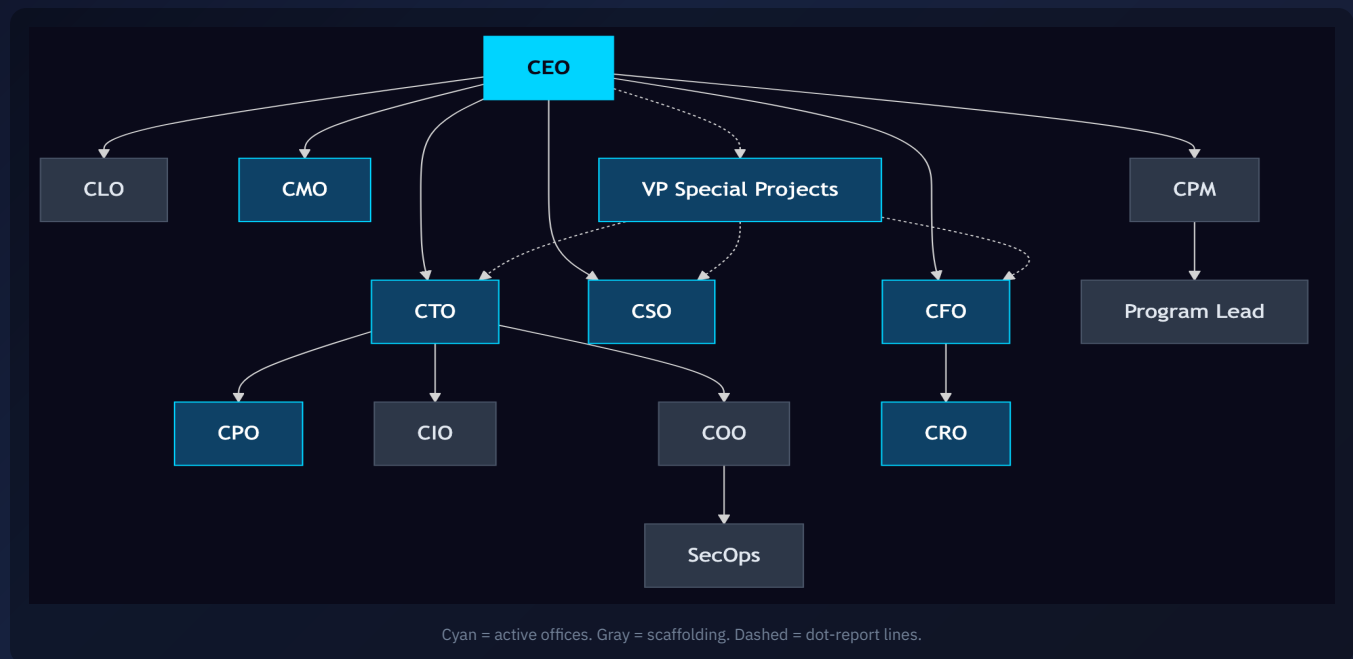


Autonomous Org Engine

Your AI agents don't need another framework. They need an office. A charter. A budget. Accountability. `drm3-office` gives every agent a persistent identity, a place in the hierarchy, and a cryptographic signature on every action they take.



What Makes This Different

AI agent frameworks are great at spinning up agents for a project. `drm3-office` asks a different question: what if the agents are the company? Persistent offices with identity, trust scores, cryptographic signatures, delegation authority, and budgets. They don't just execute tasks - they inhabit roles, build institutional memory, and report up a real hierarchy. Every action signed. Every decision traceable. The org is the product.

HOW IT WORKS

The Tick Engine

A tick is a unit of work, not time. Each office runs 11 phases. Every action signed. Every deliverable real.

- 1

Orient - Charter, SOP, memory
- 2

Inbox - Email + internal msgs
- 3

Tasks - Create from inbox
- 4

Plan - Priority sort, pick work
- 5

Deliver - Produce via LLM
- 6

Comms - Notify, ask, unblock
- 7

Resolve - 3-pass, escalate
- 8

Close - Mark tasks done
- 9

Memory - Digest, git commit
- 10

Report - Status to parent
- 11

Roll Up - Consolidate

ARCHITECTURE

Day Runner

(00:01-23:00) → Tick Engine

(leaf-to-root)

↓ Office

[charter + SOP + soul + memory + inbox + tasks]

↓ Gateway

→ claude -p (sub) | Anthropic API | OpenAI-compatible

↓ Actions

→ Write files • Send email (RustyMail MCP) • Update tasks • Git commit

↓ Graph

(SQLite) • Provenance (PROV-DM) • Receipts (Ed25519 + Merkle)

GATEWAY

PROVIDER	CONFIG	COST
Claude CLI	<code>base_url = "claude-cli"</code>	Your subscription (\$0 extra)
Anthropic API	<code>base_url = "https://api.anthropic.com"</code>	Per-token
OpenAI-compatible	<code>base_url = "https://..."</code>	Per-token



Agent email powered by RustyMail
MCP email bridge for IMAP/SMTP. Built by our friends at TexasFortress.ai

SECURITY

Path Guard Canonicalize + prefix	Email Allowlist Per-office control	Log Redaction Keys auto-stripped
SSRF Protection Blocked hosts + IPs	Ed25519 Signing Every action signed	Token Budgets Daily + monthly limits

DESIGN LINEAGE - THE DNA

drm3-admin

→ persistent offices, charters, SOPs, souls, 3-pass escalation, hierarchical reporting. The original.

drm3-avatar

→ agent identity, wallet/escrow, trust scoring, delegation, behavioral DNA.

drm3-provenance

→ Ed25519 signing, BIP39 keys, receipt chains, Merkle trees. "Cryptographic proof of AI work."

drm3-synapse

→ Elixir/OTP engine + React control plane. Graph model, API-first design.

Claude Code

→ MCP protocol, KAIROS daemon, `claude -p` gateway. Tick loop inspired by Claude's orchestration.

OpenClaw

→ model-agnostic routing, single-binary philosophy. **ruvnet** → autonomous agent patterns, MCP tool orchestration.

CrewAI

→ role-based agents, shared memory. **MetaGPT** → SOP-driven workflows. **Paperclip** → org-as-object, budget tracking.

Claude system prompts

→ charter/SOP format, structured action tags, memory digests. **Karpathy** → autoresearch, LLM OS. Parallel evolution.

EXAMPLE DAY

What a Work Day Looks Like

The daemon starts at 00:01. Each tick, every active office runs 11 phases. Here's a typical day.

00:01 • Day Opens

Day runner writes `day_open` to graph. Budget checks pass. Tick counter resets.

00:02 • Tick 1 - Leaf offices

CPO **CRO** **SecOps** **CMO** Load charter + SOP + memory. Read inbox. Pick highest-priority task. CPO drafts product spec. CRO updates revenue model. All deliverables are real files, git-committed, receipt-signed.

00:15 • Tick 1 - Mid-level

CTO **CFO** **CSO** CTO reads CPO's spec from `_inbox`. Reviews, sends feedback. CFO consolidates CRO's update. CSO runs security review. Messages flow up the tree.

00:25 • Tick 1 - CEO consolidates

CEO Reads all reports. Makes decisions. Sends board digest. Updates memory. Git commit.

00:26+ • Ticks 2, 3, 4...

Before 23:00 and under budget? Start another tick. Each builds on the last - tasks progress, memory grows, trust scores update.

09:00 • Daily Standup (cadence)

CEO **CTO** **CSO** **CFO** **CMO** Async status: blockers, priorities, updates. Meeting artifact recorded in graph.

23:00 • Day Closes

No new ticks. In-flight drain. Day summary to graph. Memory digests committed. Ready for tomorrow.

COMING FROM AVATAR

Trust Scoring

Completion rate, response time, escalation rate, deliverable quality, budget adherence. Weighted composite score per office. Computed from graph events.

Delegation + Escrow

<delegate> action tag. Budget allocation/spend/escrow tracking. Accept/reject/complete/dispute lifecycle. Amazon-style department budgets.

State Deriver

Office mood (productive/stressed/idle/blocked), current focus, workload, relationship strength. All computed from graph events, never stored.

Deep Persona (7-table)

Beliefs, constraints (hard/soft), styles, biases, habits, knowledge domains. TOML per office, injected into the LLM context automatically.

drm3-provenance: Cryptographic Proof of AI Work

Every action produces a signed `Receipt`: Ed25519 over `input_hash + output_hash + timestamp + cost + model`. Receipts chain via `parent_id` into a Merkle tree per tick. One BIP39 mnemonic → one keyring → deterministic per-office keys. W3C PROV-DM: `Entity`, `Activity`, `Agent`, `wasGeneratedBy`, `wasDerivedFrom`. The unforgeable audit trail for everything your AI org does.

WHAT EMERGES FROM THE ARCHITECTURE

Accountability

Persistent Ed25519 identity + signed receipts + Merkle chains + queryable graph = every action attributable, every chain verifiable. Not a feature you bolt on — it's the substrate everything runs on.

Governance

Hierarchy + delegation + escrow + trust scoring = organizational economics. Offices delegate budget, track completion, build trust over time. The system has opinions about who's performing.

Institutional Memory

Day runner + tick loop + memory digests + git commits = the org remembers. Not conversation-scoped. Career-scoped. Every day, a digest. Every tick, a commit. Every decision, queryable.

The Compound Effect

Any one of these is a weekend project. Together they create something none of them produce alone: an autonomous organization that can be audited, governed, and trusted. That combination is the architecture.

Identity, Provenance, and Trust

Every office has a cryptographic identity. Every action is signed. Every chain is verifiable.

Key Derivation

BIP39 Mnemonic (one per org)
| HKDF derive per-office keypair
|- path: **agent:ceo** - Ed25519
|- path: **agent:cto** - Ed25519
|- path: **agent:*** - Ed25519
Deterministic. Recoverable.

Receipt Chain

Tick Start - LLM Call - sign(prompt, response)
| Deliverable - sign(path, hash)
| Message - sign(to, subject)
| Merkle - root = hash(r1||r2|...)
Tick End - store in **tick_chains**

W3C PROV-DM

Entity document, decision, message | **Activity** tick, phase, LLM call |
Agent office, model, human
wasGeneratedBy | wasDerivedFrom | wasAttributedTo

Security

Path traversal guard | Email allowlist per office | Log redaction (sk-ant-, ghp_, Bearer) | SSRF protection (localhost, metadata, private IPs) | Token budgets (daily + monthly) | AES-256-GCM credential vault

30 MODULES

engine Tick + receipts	gateway 3-provider LLM	graph SQLite + edges	office Charter, SOP, DNA
sync FS + git + Plane	comms IMAP/SMTP + MCP	telemetry Logs, PIDs, usage	provenance W3C PROV-DM
identity Ed25519 keyring	trust Composite scoring	delegation Escrow + budgets	state Derived state
persona 7-table behavioral	credentials AES-256 vault	cadence Standups, meetings	browser Fetch + SSRF
knowledge Ask + search	viz Graphviz + brand	tui Interactive terminal	scheduler Cron daemon
memory Digests	mcp Tool servers	sink BigQuery export	errors Typed hierarchy